



Saisaran M 251801281-2025-AIDS ▾

**S2****Started on** Thursday, 24 September 2026, 10:32 AM**State** Finished**Completed on** Thursday, 24 September 2026, 11:02 AM**Time taken** 29 mins 31 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

**Question 1** | Correct | Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of  $n$  integers between 1 and  $n$ , find one number that repeats.

Input Format:

First Line - Number of elements

$n$  Lines -  $n$  Elements

Output Format:

Element  $x$  - That is repeated

**For example:**

| Input          | Result |
|----------------|--------|
| 5<br>1 1 2 3 4 | 1      |

**Answer:** (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main() {
4      int n;
5      scanf("%d", &n);
6
7      int arr[n];
8
9      for (int i = 0; i < n; i++) {
10         scanf("%d", &arr[i]);
11     }
12
13     for (int i = 0; i < n; i++) {
14         for (int j = i + 1; j < n; j++) {
15             if (arr[i] == arr[j]) {
16                 printf("%d\n", arr[i]);
17                 return 0;
18             }
19         }
20     }
21
22     return 0;
23 }
24
25

```

|   | Input                        | Expected | Got |   |
|---|------------------------------|----------|-----|---|
| ✓ | 11<br>10 9 7 6 5 1 2 3 8 4 7 | 7        | 7   | ✓ |
| ✓ | 5<br>1 2 3 4 4               | 4        | 4   | ✓ |
| ✓ | 5<br>1 1 2 3 4               | 1        | 1   | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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**Saisaran M 251801281-2025-AIDS** ▾**S2****Started on** Thursday, 24 September 2026, 11:02 AM**State** Finished**Completed on** Thursday, 24 September 2026, 11:33 AM**Time taken** 31 mins 24 secs**Marks** 1.00/1.00**Grade** **10.00** out of 10.00 (**100%**)

**Question 1** | Correct | Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

**For example:**

| Input          | Result |
|----------------|--------|
| 5<br>1 1 2 3 4 | 1      |

**Answer:** (penalty regime: 0 %)

```

1  #include <stdio.h>
2  #include <stdlib.h>
3
4  int findDuplicate(const int* nums, int n) {
5
6      int tortoise = nums[0];
7      int hare = nums[0];
8
9      do {
10         tortoise = nums[tortoise];
11         hare = nums[nums[hare]];
12     } while (tortoise != hare);
13
14
15     tortoise = nums[0];
16     while (tortoise != hare) {
17         tortoise = nums[tortoise];
18         hare = nums[hare];
19     }
20
21     return tortoise;
22 }
23
24 int main() {
25     int n;
26
27     // Read the number of elements
28     if (scanf("%d", &n) != 1) {
29         return 1;
30     }
31
32     // Dynamically allocate memory for n elements
33     int* nums = (int*)malloc(n * sizeof(int));
34     if (nums == NULL) {
35         return 1;
36     }
37
38     // Read the array elements
39     for (int i = 0; i < n; i++) {
40         if (scanf("%d", &nums[i]) != 1) {
41             free(nums);
42             return 1;
43         }
44     }
45
46     // Find and print the duplicate
47     int duplicate = findDuplicate(nums, n);
48     printf("%d\n", duplicate);
49
50     // Free allocated memory
51     free(nums);
52

```

|   | Input                        | Expected | Got |   |
|---|------------------------------|----------|-----|---|
| ✓ | 11<br>10 9 7 6 5 1 2 3 8 4 7 | 7        | 7   | ✓ |
| ✓ | 5<br>1 2 3 4 4               | 4        | 4   | ✓ |
| ✓ | 5<br>1 1 2 3 4               | 1        | 1   | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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[Saisaran M 251801281-2025-AIDS](#) ▾**S2****Started on** Thursday, 24 September 2026, 11:35 AM**State** Finished**Completed on** Thursday, 8 October 2026, 10:31 AM**Time taken** 13 days 22 hours**Marks** 1.00/1.00**Grade** **10.00** out of 10.00 (**100%**)

**Question 1** | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

**For example:**

| Input                                    | Result |
|------------------------------------------|--------|
| 1<br>3 10 17 57<br>6<br>2 7 10 15 57 246 | 10 57  |

**Answer:** (penalty regime: 0 %)

```

1 #include <stdio.h>
2
3 int main() {
4     int T;
5     scanf("%d", &T);
6
7     while (T--) {
8         int n1, n2;
9
10        scanf("%d", &n1);
11        int a[n1];
12
13        for (int i = 0; i < n1; i++) {
14            scanf("%d", &a[i]);
15        }
16
17        scanf("%d", &n2);
18        int b[n2];
19
20        for (int i = 0; i < n2; i++) {
21            scanf("%d", &b[i]);
22        }

```

```
23
24
25 int i = 0, j = 0;
26 int first = 1;
27
28 while (i < n1 && j < n2) {
29     if (a[i] == b[j]) {
30         if (!first)
31             printf(" ");
32         printf("%d", a[i]);
33         first = 0;
34
35         i++;
36         j++;
37     }
38     else if (a[i] < b[j]) {
39         i++;
40     }
41     else {
42         j++;
43     }
44 }
45
46 printf("\n");
47 }
48
49 return 0;
50 }
```

|   | Input                                    | Expected | Got   |   |
|---|------------------------------------------|----------|-------|---|
| ✓ | 1<br>3 10 17 57<br>6<br>2 7 10 15 57 246 | 10 57    | 10 57 | ✓ |
| ✓ | 1<br>6 1 2 3 4 5 6<br>2<br>1 6           | 1 6      | 1 6   | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.



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[Saisaran M 251801281-2025-AIDS](#) ▾**S2****Started on** Thursday, 8 October 2026, 10:32 AM**State** Finished**Completed on** Thursday, 8 October 2026, 10:32 AM**Time taken** 40 secs**Marks** 1.00/1.00**Grade** **10.00** out of 10.00 (**100%**)

**Question 1** | Correct | Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

**For example:**

| Input                                    | Result |
|------------------------------------------|--------|
| 1<br>3 10 17 57<br>6<br>2 7 10 15 57 246 | 10 57  |

**Answer:** (penalty regime: 0 %)

```

1 #include <stdio.h>
2
3 int main() {
4     int T;
5     scanf("%d", &T);
6
7     while (T--) {
8         int n1, n2;
9
10        scanf("%d", &n1);
11
12        int a[n1];
13        for (int i = 0; i < n1; i++) {
14            scanf("%d", &a[i]);
15        }
16
17        scanf("%d", &n2);
18
19        int b[n2];
20        for (int i = 0; i < n2; i++) {
21            scanf("%d", &b[i]);
22        }

```

```

23     ,
24     int i = 0, j = 0;
25
26     while (i < n1 && j < n2) {
27         if (a[i] == b[j]) {
28             printf("%d ", a[i]);
29             i++;
30             j++;
31         }
32         else if (a[i] < b[j]) {
33             i++;
34         }
35         else {
36             j++;
37         }
38     }
39
40     printf("\n");
41 }
42
43     return 0;
44 }

```

|   | Input                                    | Expected | Got   |   |
|---|------------------------------------------|----------|-------|---|
| ✓ | 1<br>3 10 17 57<br>6<br>2 7 10 15 57 246 | 10 57    | 10 57 | ✓ |
| ✓ | 1<br>6 1 2 3 4 5 6<br>2<br>1 6           | 1 6      | 1 6   | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.



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Saisaran M 251801281-2025-AIDS ▾

S2**Started on** Thursday, 8 October 2026, 10:33 AM**State** Finished**Completed on** Thursday, 8 October 2026, 10:33 AM**Time taken** 43 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

**Question 1** | Correct | Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that  $A[j] - A[i] = k$ ,  $i \neq j$ .

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as  $5 - 1 = 4$

So Return 1.

**For example:**

| Input           | Result |
|-----------------|--------|
| 3<br>1 3 5<br>4 | 1      |

**Answer:** (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main() {
4      int n, k;
5
6      scanf("%d", &n);
7
8      int a[n];
9
10     for (int i = 0; i < n; i++) {
11         scanf("%d", &a[i]);
12     }
13
14     scanf("%d", &k);
15
16     int i = 0, j = 1;
17
18     while (i < n && j < n) {
19         if (i == j) {
20             j++;
21         }
22         else if (a[j] - a[i] == k) {
23             printf("1\n");
24             return 0;
25         }
26         else if (a[j] - a[i] < k) {
27             j++;
28         }
29         else {
30             i++;
31         }
32     }
33
34     printf("0\n");
35
36     return 0;
37 }
```

|   | Input                                 | Expected | Got |   |
|---|---------------------------------------|----------|-----|---|
| ✓ | 3<br>1 3 5<br>4                       | 1        | 1   | ✓ |
| ✓ | 10<br>1 4 6 8 12 14 15 20 21 25<br>1  | 1        | 1   | ✓ |
| ✓ | 10<br>1 2 3 5 11 14 16 24 28 29<br>0  | 0        | 0   | ✓ |
| ✓ | 10<br>0 2 3 7 13 14 15 20 24 25<br>10 | 1        | 1   | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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Saisaran M 251801281-2025-AIDS ▾

**S2****Started on** Thursday, 8 October 2026, 11:16 AM**State** Finished**Completed on** Thursday, 8 October 2026, 11:18 AM**Time taken** 2 mins 24 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

**Question 1** | Correct | Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that  $A[j] - A[i] = k$ ,  $i \neq j$ .

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as  $5 - 1 = 4$

So Return 1.

**For example:**

| Input           | Result |
|-----------------|--------|
| 3<br>1 3 5<br>4 | 1      |

**Answer:** (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main() {
4      int n;
5      scanf("%d", &n);
6
7      int A[n];
8
9      for (int i = 0; i < n; i++) {
10         scanf("%d", &A[i]);
11     }
12
13     int k;
14     scanf("%d", &k);
15
16     int i = 0, j = 1;
17
18     while (i < n && j < n) {
19         if (i == j) {
20             j++;
21             continue;
22         }
23
24         int diff = A[j] - A[i];
25
26         if (diff == k) {
27             printf("1");
28             return 0;
29         }
30         else if (diff < k) {
31             j++;
32         }
33         else {
34             i++;
35         }
36     }
37
38     printf("0");
39

```

```
40 | return 0;  
41 | }
```

|   | Input                                 | Expected | Got |   |
|---|---------------------------------------|----------|-----|---|
| ✓ | 3<br>1 3 5<br>4                       | 1        | 1   | ✓ |
| ✓ | 10<br>1 4 6 8 12 14 15 20 21 25<br>1  | 1        | 1   | ✓ |
| ✓ | 10<br>1 2 3 5 11 14 16 24 28 29<br>0  | 0        | 0   | ✓ |
| ✓ | 10<br>0 2 3 7 13 14 15 20 24 25<br>10 | 1        | 1   | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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